Statistics for Computer Scientists

Lecture 7: Statistical testing

31 October 2025

Universiteit Leiden
The Netherlands

Where are we

Lecture	Topic	Book
5 Sept	Introduction, statistics fundamentals	Chapter 1
12 Sept	Descriptive statistics	Chapter 2
19 Sept	Bivariate relationships	Chapter 3
26 Sept	Data collection	Chapter 4
10 Oct	Sampling distributions	Chapter 7
17 Oct	Estimation	Chapter 8
31 Oct	Significance testing	Chapter 9
7 Nov	Comparing two groups	Chapter 10
14 Nov	Categorical data analysis	Chapter 11
21 Nov	Linear regression	Chapter 12
28 Nov	Multiple regression	Chapter 13
5 Dec	Tba	
12 Dec	Tba	

Today's lecture

- Recap lecture 6 (estimation)

- Statistical hypothesis testing
  - Hypotheses and tests
  - Testing a single mean
  - Testing a single proportion
  - Decisions and errors

Recap estimation

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Statistical inference?

Point estimation

Three common **point estimators**:

- **Sample mean** estimates population mean: $\hat{\mu} = \bar{y} = \frac{\sum y_i}{n}$
- **Sample standard deviation** estimates population sd: $\hat{\sigma} = s = \sqrt{\frac{\sum(y_i - \bar{y})^2}{n-1}}$
- **Sample proportion** estimates population proportion: $\hat{\pi}(= p) = \frac{\#successes}{n}$

Interval estimation

- A **confidence interval** (CI) is an interval of numbers believed to contain the parameter value
- The **confidence level** is the probability that the interval contains the parameter (typically a number close to 1 *(or 100%)*, such as 0.95 or 0.99 *(95% or 99%)*)
- CIs are of the form **point estimate ± margin of error**
- Margin of error is based on the **standard error** and a **score**, depending on confidence level

Hypotheses and tests

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Significance tests

Goal: use statistical methods to check hypotheses, such as:

- "Mental health tends to be better at higher levels of socio-economic status"
  - (an effect)

- "For treating anorexia, cognitive behavioural and family therapies have the same effect"
  - (no effect)

Hypotheses and tests

- **Hypothesis**: a prediction about a population, expressed in terms of parameters for given variables

- **Significance test**: uses data to summarize evidence about a hypothesis by comparing sample estimates of parameters to values predicted by the hypothesis

- We answer questions such as: "if the (null) hypothesis were true, how (un)likely would it be to get the estimate we obtained?"

Five parts of a significance test

1. Assumptions

- Type of data (quantitative, categorical)
- Sampling method (random?)
- Population distribution (normal?)
- Sample size (large?)
- Possibly others

2. Hypotheses

- Null hypothesis (H_0): states that parameter takes a specific value (usually: “no effect”)
- Alternative hypothesis (H_a): states that parameter falls in an alternative range of values (“effect”)

Five parts of a significance test

3. Test statistic

- Compares data to H_0, often by finding the *number of standard errors* between the sample estimate and the H_0-value of the parameter

4. P-value

- Probability of finding the observed value for the test statistic, or even more extreme, *if H_0 would be true*
- The smaller the P-value, the stronger the evidence **against** H_0

Five parts of a significance test

5. Conclusion

- If no decision is needed: report and *interpret* the P-value
- If yes/no decision is needed: reject H_0 is P-value is below a cut-off (α, usually 0.05 or 0.01), which was selected **in advance** (ideally before data collection and certainly before any calculations!)

- *Always relate the P-value to the context*

About (statistical) significance

- A test result is significant if $P \leq \alpha$
  - Most common: $\alpha = 0.05$
  - Meaning that if H_0 were true, the probability of finding this sample estimate is less than 5%
- If the P-value is not small enough, we fail to reject H_0
  (we have not proven it is true, only that it is plausible)

- Only two options:
  - $P \leq \alpha$: reject H_0
  - $P > \alpha$: do not reject H_0
- **H_0 can never be accepted; only rejected or failed to reject!**

Testing a single mean

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Significance test for mean

1. Assumptions

- We have a (single) **quantitative variable**
- Data are obtained using **randomization**
- The population distribution is approximately **normal**

2. Hypotheses

- $H_0: \mu = \mu_0$ where μ_0 is a particular value for the population mean, usually "no effect"
- $H_a: \mu \neq \mu_0$ two-sided alternative, includes both > and < null value

Significance test for mean

3. Test statistic

- For this test, the test statistic t is the number of standard errors that the sample falls from the H_0 value (μ_0)

$$t = \frac{\bar{y} - \mu_0}{se}, \qquad se = \frac{s}{\sqrt{n}}$$

- When H_0 is true, the **sampling distribution** of the test statistic t is the t-distribution with $df = n - 1$

Unsurprisingly, this is called the t-test

Significance test for mean

4. P-value

- The p-value is the probability that we find our observed value or more extreme (further away from μ_0), **assuming that H_0 is true**.
- So, small p-value means that this observed value is very unlikely if H_0 is true; therefore H_0 is likely to be false
- P-value $= P(|T| \geq |t| \ \mid \ H_0)$

5. Conclusion

- Report and interpret the p-value. If needed, make a decision about H_0. Always use context!

Example: weight gain after therapy

- Anorexia study: weight gain after family therapy measured in 17 girls

- X = pounds gained
- Sample data:
  11.4, 11.0, 5.5, 9.4, 13.6, -2.9, -0.1, 7.4, 21.5, -5.3, -3.8, 13.4, 13.1, 9.0, 3.9, 5.7, 10.7

Five steps!

1. Assumptions
2. Hypotheses
3. Test statistic
4. P-value
5. Conclusion

1. Assumptions

- Quantitative variable?
  - Yes, number of pounds gained
- Data obtained using randomization?
  - Not mentioned, assume yes
- Normal population distribution
  - Not mentioned, assume yes

2. Hypotheses

- In words: “does the treatment have an effect?”
- How to formulate this in a statistical way? Be specific!
  Always use a **parameter** and a **value**

- Let μ be the population mean weight change
- Then:
  - $H_0: \mu_0 = 0$ (no effect)
  - $H_a: \mu_0 \neq 0$ (effect)

3. Test statistic

- We are performing a t-test, so:

$$t = \frac{\bar{y} - \mu_0}{se}, \qquad se = \frac{s}{\sqrt{n}}$$

- Calculate $\bar{y}$ and s from data, then calculate se, then calculate t

$$\bar{y} = 7.265, \qquad s = 7.157$$

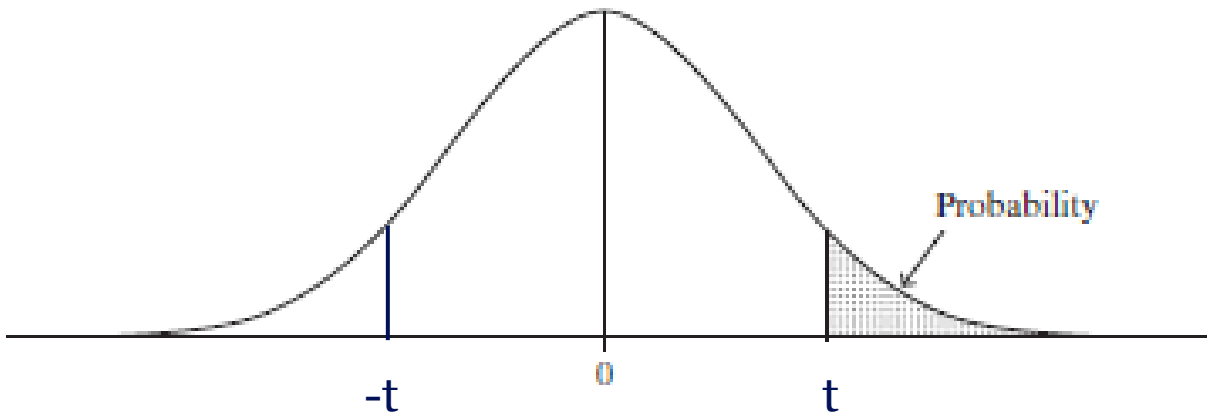
$$se = \frac{7.157}{\sqrt{17}} = 1.736$$

$$t = \frac{7.264 - 0}{1.736} = 4.2$$

4. P-value

- P is the probability that the test statistic equals the observed value, or is even more extreme. In the case of $t = 4.2$, more extreme means $t > 4.2$ or $t < -4.2$

TABLE B: *t* Distribution Critical Values



4. P-value

- We can find the p-value using Table B
- $df = n - 1 = 16$ gives the correct row

- But which column?

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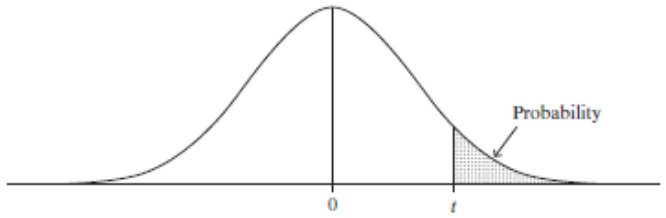

	Confidence Level					
	80%	90%	95%	98%	99%	99.8%
	Right-Tail Probability					
df	$t_{.100}$	$t_{.050}$	$t_{.025}$	$t_{.010}$	$t_{.005}$	$t_{.001}$
1	3.078	6.314	12.706	31.821	63.656	318.289
2	1.886	2.920	4.303	6.965	9.925	22.328
3	1.638	2.353	3.182	4.541	5.841	10.214
4	1.533	2.132	2.776	3.747	4.604	7.173
5	1.476	2.015	2.571	3.365	4.032	5.894
6	1.440	1.943	2.447	3.143	3.707	5.208
7	1.415	1.895	2.365	2.998	3.499	4.785
8	1.397	1.860	2.306	2.896	3.355	4.501
9	1.383	1.833	2.262	2.821	3.250	4.297
10	1.372	1.812	2.228	2.764	3.169	4.144
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23	1.319	1.714	2.069	2.500	2.807	3.485
24	1.318	1.711	2.064	2.492	2.797	3.467
25	1.316	1.708	2.060	2.485	2.787	3.450
26	1.315	1.706	2.056	2.479	2.779	3.435
27	1.314	1.703	2.052	2.473	2.771	3.421
28	1.313	1.701	2.048	2.467	2.763	3.408
29	1.311	1.699	2.045	2.462	2.756	3.396
30	1.310	1.697	2.042	2.457	2.750	3.385
40	1.303	1.684	2.021	2.423	2.704	3.307
50	1.299	1.676	2.009	2.403	2.678	3.261
60	1.296	1.671	2.000	2.390	2.660	3.232
80	1.292	1.664	1.990	2.374	2.639	3.195
100	1.290	1.660	1.984	2.364	2.626	3.174
∞	1.282	1.645	1.960	2.326	2.576	3.091

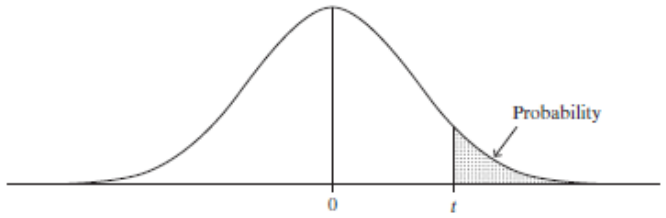
Source: "Table of Percentage Points of the *t*-Distribution." Computed by Maxine Merrington, Biometrika, 32 (1941): 300. Reproduced by permission of the Biometrika trustees.

4. P-value

- Table uses right-tail probability, we need both tails

- Solution: use column of $t_{\alpha/2}$
- If $\alpha = 0.05$, the test is significant if the absolute value of our calculated t exceeds the critical value of 2.120

TABLE B: *t* Distribution Critical Values



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4. P-value

- At an α-level of 0.05, the critical value of t for significance is 2.120. The observed test statistic is 4.2, which exceeds the critical value
- Therefore, we know that P < 0.05

(We can also see that the observed t exceeds the critical value $t_{0.001}$, so we even know that $P < 0.002$!)

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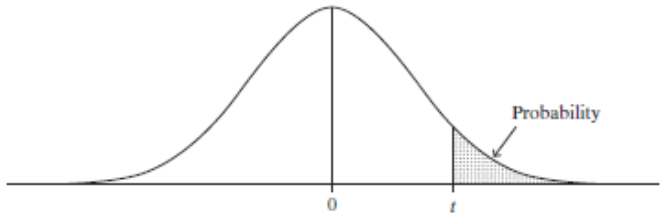

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5. Conclusion

$P < 0.05$, therefore we **reject** the null-hypothesis that the mean weight change in the population is 0. There is an effect of the treatment.

Exact p-values?

Exact p-values typically cannot be calculated by hand and won't be needed in the exam

You can use JASP and the book's 'web apps' to compute exact p-values

Significance test and confidence interval

- When P-value ≤ 0.05 in two-sided test, 95% CI for μ **does not contain** H_0 value of μ.

- When P-value > 0.05 in two-sided test, 95% CI necessarily **contains** H_0 value of μ.

This is true for "two-sided" tests!

CI gives more information about actual value of μ than P-value

Two-sided?

- "Does the treatment have an effect?"
  - $H_0: \mu_0 = 0$
  - $H_a: \mu_0 \neq 0$

Two-sided

- "Does the treatment have a positive effect?"
  - $H_0: \mu_0 = 0$
  - $H_a: \mu_0 > 0$

One-sided

One-sided test about mean

- **Same example**: If study predicts family therapy has *positive* effect, could use $\mathrm{H_a}: \mu > 0$ (we are hoping for weight gain)

- In this case, we only consider positive t-values, so only the right-tail probability!
- Calculation of the test statistic (step 3) **does not change**: $t = 4.2$

4. P-value

- Table uses right-tail probability, this time, we do **not** need both tails

- Solution: use column of t_α, rather than $t_{\alpha/2}$

- If $\alpha = 0.05$, the test is significant if the absolute value of our calculated t exceeds the critical value of 1.746 (rather than 2.120)

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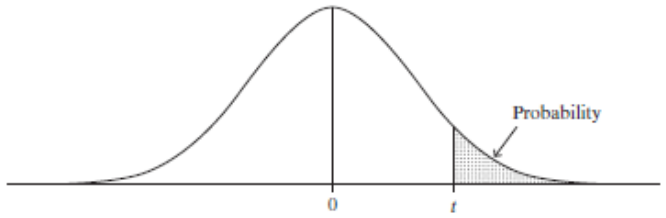

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Two-sided vs one-sided

T Score: 4.2
DF: 16

Significance Level:

- ○ .01
- ◉ .05
- ○ .10

One-tailed or two-tailed hypothesis?:

- ○ One-tailed
- ◉ Two-tailed

The *p*-value is .000678.

The result is significant at $p < .05$.

T Score: 4.2
DF: 16

Significance Level:

- ○ .01
- ◉ .05
- ○ .10

One-tailed or two-tailed hypothesis?:

- ◉ One-tailed
- ○ Two-tailed

The *p*-value is .000339.

The result is significant at $p < .05$.

- P-value for two-sided test is twice the p-value for one-sided test
- It is easier to find a significant result using a one-sided test!
  Why not always use this?

Effect of sample size

- CLT and robustness
  - With large n (> 30), assumption of normal population distribution not important because of Central Limit Theorem
  - For small n, the *two-sided t* test is robust against violations of that assumption. One-sided test is *not* robust!

- For given observed sample mean and standard deviation, the larger the sample size n, the larger the test statistic (because *se* in denominator is smaller) and the smaller the P-value. That is, **more data = more evidence**!
  - We're more likely to reject a false H_0 when we have a larger sample size (the test then has more "power")
  - With large n, "statistical significance" not the same as "practical significance"

Testing a single proportion

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Same five steps!

1. Assumptions
2. Hypotheses
3. Test statistic
4. P-value
5. Conclusion

Significance test for proportion

1. Assumptions

- We have a **categorical** variable
- Data are obtained using **randomization**
- The sample size is sufficiently large, i.e., the sample (of given sample size) is expected to contain **at least 15 successes and failures under the null hypothesis**

2. Hypotheses

- $H_0: \pi = \pi_0$ where π_0 is a particular value for the population proportion
- $H_a: \pi \neq \pi_0$ two-sided
- or $H_a: \pi < \pi_0$ one-sided
- or $H_a: \pi > \pi_0$ one-sided

Note that the book uses p and p_0 instead of π and π_0, but for clarity we more clearly distinguish parameters from statistics. Notation is not universal.

Significance test for proportion

3. Test statistic

- For this test, the test statistic z is the number of standard errors that the sample falls from the H_0 value (π_0)

$$z = \frac{\hat{\pi} - \pi_0}{\sigma_{\hat{\pi}}}, \qquad \sigma_{\hat{\pi}} = \mathrm{se}_0 = \sqrt{\pi_0(1-\pi_0)/n}$$

When H_0 is true, the **sampling distribution** of the test statistic z follows the standard normal distribution ($\mu = 0, \quad \sigma = 1$)

- **This is also called the z-test**

Careful! This is different from the standard error for a confidence interval, which is:
$se = \sqrt{\hat{\pi}(1-\hat{\pi})/n}$

Significance test for proportion

4. P-value

- P-value = $P(|Z| \geq |z| \ \mid \ H_0)$
- Now instead of using Table B, we have to use Table A, because these are z-values!

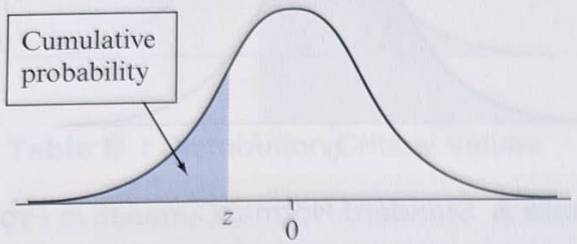
5. Conclusion

- Report and interpret the p-value. If needed, make a decision about H_0. Always use context!

4. P-value

- Use Table A to get p-value

- Pay attention! Table A gives left-tail probabilities. What if we have a two-sided alternative hypothesis?
- Double the left-tail probability!
- And what if we need a right-tail probability?



The cumulative probability for z is the area under the standard normal curve to the left of z

Table A Standard Normal Cumulative Probabilities

z	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
−3.4	.0003	.0003	.0003	.0003	.0003	.0003	.0003	.0003	.0003	.0002
−3.3	.0005	.0005	.0005	.0004	.0004	.0004	.0004	.0004	.0004	.0003
−3.2	.0007	.0007	.0006	.0006	.0006	.0006	.0006	.0005	.0005	.0005
−3.1	.0010	.0009	.0009	.0009	.0008	.0008	.0008	.0008	.0007	.0007
−3.0	.0013	.0013	.0013	.0012	.0012	.0011	.0011	.0011	.0010	.0010
−2.9	.0019	.0018	.0018	.0017	.0016	.0016	.0015	.0015	.0014	.0014
−2.8	.0026	.0025	.0024	.0023	.0023	.0022	.0021	.0021	.0020	.0019
−2.7	.0035	.0034	.0033	.0032	.0031	.0030	.0029	.0028	.0027	.0026
−2.6	.0047	.0045	.0044	.0043	.0041	.0040	.0039	.0038	.0037	.0036
−2.5	.0062	.0060	.0059	.0057	.0055	.0054	.0052	.0051	.0049	.0048
−2.4	.0082	.0080	.0078	.0075	.0073	.0071	.0069	.0068	.0066	.0064
−2.3	.0107	.0104	.0102	.0099	.0096	.0094	.0091	.0089	.0087	.0084
−2.2	.0139	.0136	.0132	.0129	.0125	.0122	.0119	.0116	.0113	.0110
−2.1	.0179	.0174	.0170	.0166	.0162	.0158	.0154	.0150	.0146	.0143
−2.0	.0228	.0222	.0217	.0212	.0207	.0202	.0197	.0192	.0188	.0183

Classified as Internal | Intern

Example: can dogs smell cancer?

- From the British Medical Journal, 25 September 2004
- Can dogs detect cancer in urine samples? In each trial, one urine sample of a patient with bladder cancer was placed among six control urine samples
- Do dogs make the correct selection more often than expected with random guessing?
- Result: in 54 trials, dogs made the correct selection 22 times

1. Assumptions

- Categorical variable?
  - Yes, categorical variable with possible values “correct selection” and “incorrect selection”

- Data obtained using randomization?
  - Not mentioned, we have to assume this

- >15 observations in each category expected?
  - **No**: $n\pi_0 = 54\left(\frac{1}{7}\right) = 7.7$

2. Hypotheses

- Let π be the population probability that the correct sample (out of 7 total) is chosen

- If dogs cannot smell cancer, we expect them to randomly select the correct sample 1/7 of the time

- $H_0: \pi = \frac{1}{7} = 0.143$
- $H_a: \pi > 0.143$ one-sided alternative

3. Test statistic

$$se_0 = \sqrt{\frac{\pi_0(1-\pi_0)}{n}} = \sqrt{\frac{\frac{1}{7} * \frac{6}{7}}{54}} = 0.0476$$

$$z = \frac{\hat{\pi} - \pi_0}{se_0} = \frac{\frac{22}{54} - \frac{1}{7}}{0.0476} = 5.6$$

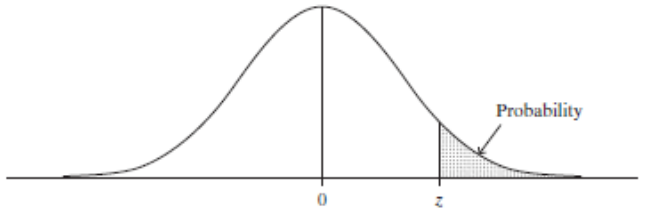
So, our observed proportion is 5.6 standard errors from the null-hypothesis proportion! Can you guess the significance already?

4. P-value

- For z=5.6, the left-tail probability is >0.999999713, so the right-tail probability is <0.000000287

- This is the correct p-value, because we have a one-sided H_a
- If we were testing two-sided, we would multiply the value by 2

TABLE A: Normal curve tail probabilities. Standard normal probability in right-hand tail (for negative values of z, probabilities are found by symmetry).



	Second Decimal Place of z									
z	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
0.0	.5000	.4960	.4920	.4880	.4840	.4801	.4761	.4721	.4681	.4641
0.1	.4602	.4562	.4522	.4483	.4443	.4404	.4364	.4325	.4286	.4247
0.2	.4207	.4168	.4129	.4090	.4052	.4013	.3974	.3936	.3897	.3859
0.3	.3821	.3783	.3745	.3707	.3669	.3632	.3594	.3557	.3520	.3483
0.4	.3446	.3409	.3372	.3336	.3300	.3264	.3228	.3192	.3156	.3121
0.5	.3085	.3050	.3015	.2981	.2946	.2912	.2877	.2843	.2810	.2776
0.6	.2743	.2709	.2676	.2643	.2611	.2578	.2546	.2514	.2483	.2451
0.7	.2420	.2389	.2358	.2327	.2296	.2266	.2236	.2206	.2177	.2148
0.8	.2119	.2090	.2061	.2033	.2005	.1977	.1949	.1922	.1894	.1867
0.9	.1841	.1814	.1788	.1762	.1736	.1711	.1685	.1660	.1635	.1611
1.0	.1587	.1562	.1539	.1515	.1492	.1469	.1446	.1423	.1401	.1379
1.1	.1357	.1335	.1314	.1292	.1271	.1251	.1230	.1210	.1190	.1170
1.2	.1151	.1131	.1112	.1093	.1075	.1056	.1038	.1020	.1003	.0985
1.3	.0968	.0951	.0934	.0918	.0901	.0885	.0869	.0853	.0838	.0823
1.4	.0808	.0793	.0778	.0764	.0749	.0735	.0722	.0708	.0694	.0681
1.5	.0668	.0655	.0643	.0630	.0618	.0606	.0594	.0582	.0571	.0559
1.6	.0548	.0537	.0526	.0516	.0505	.0495	.0485	.0475	.0465	.0455
1.7	.0446	.0436	.0427	.0418	.0409	.0401	.0392	.0384	.0375	.0367
1.8	.0359	.0352	.0344	.0336	.0329	.0322	.0314	.0307	.0301	.0294
1.9	.0287	.0281	.0274	.0268	.0262	.0256	.0250	.0244	.0239	.0233
2.0	.0228	.0222	.0217	.0212	.0207	.0202	.0197	.0192	.0188	.0183
2.1	.0179	.0174	.0170	.0166	.0162	.0158	.0154	.0150	.0146	.0143
2.2	.0139	.0136	.0132	.0129	.0125	.0122	.0119	.0116	.0113	.0110
2.3	.0107	.0104	.0102	.0099	.0096	.0094	.0091	.0089	.0087	.0084
2.4	.0082	.0080	.0078	.0075	.0073	.0071	.0069	.0068	.0066	.0064
2.5	.0062	.0060	.0059	.0057	.0055	.0054	.0052	.0051	.0049	.0048
2.6	.0047	.0045	.0044	.0043	.0041	.0040	.0039	.0038	.0037	.0036
2.7	.0035	.0034	.0033	.0032	.0031	.0030	.0029	.0028	.0027	.0026
2.8	.0026	.0025	.0024	.0023	.0023	.0022	.0021	.0021	.0020	.0019
2.9	.0019	.0018	.0017	.0017	.0016	.0016	.0015	.0015	.0014	.0014
3.0	.00135									
3.5	.000233									
4.0	.0000317									
5.0	.000000287									

Source: R. E. Walpole, *Introduction to Statistics* (New York: Macmillan, 1968).

5. Conclusion

- At an α-level of 0.05 (or in this case, even 0.01, or 0.001), the observed proportion of correct answers is significantly higher than 1/7

- Therefore, we reject the null-hypothesis, and conclude that the dogs' selections are better than random guessing!
  - But note that one of the test's assumptions was not satisfied, which may impact the outcome

Decisions and errors

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Decisions in tests

- ***α*-level**: the pre-specified "hurdle" for which the null hypothesis will be rejected, if the P-value falls below it

P-Value	H_0 Conclusion	H_a Conclusion
$\leq .05$	Reject	Accept
$> .05$	Do not Reject	Do not Accept

- **Rejection region**: values of the test statistic for which we reject the null hypothesis
  - For two-sided z-tests with $\alpha = 0.05$, the rejection region is $|z| > 1.96$

Two types of error

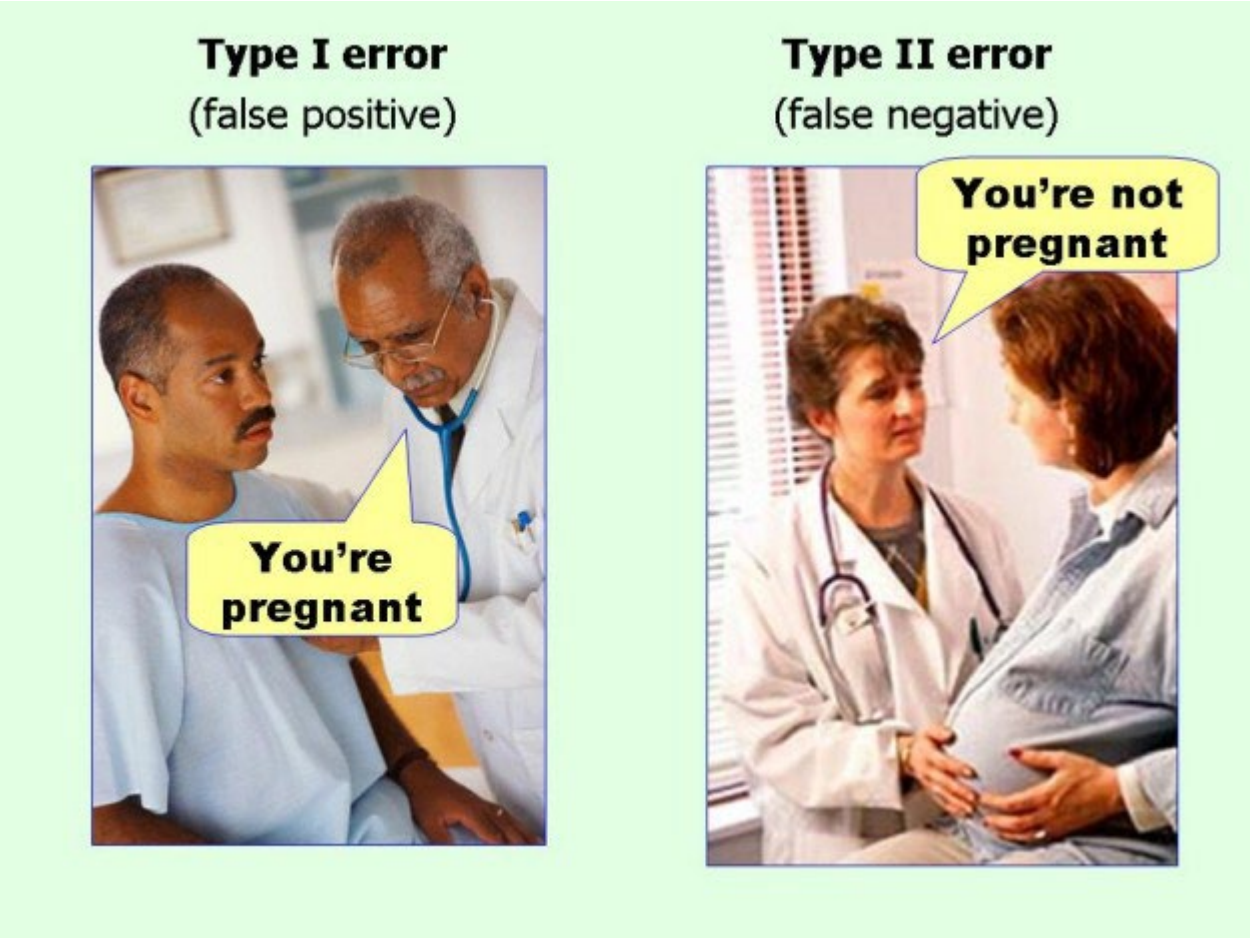
- Type I error: reject H_0 when it is true
- Type II error: fail to reject H_0 when it is false

	$P \leq \alpha$	$P > \alpha$
H_0 is true	Type I error	All good
H_0 is false	All good	Type II error

Example

H_0: you are not pregnant ("no effect")

H_a: you are pregnant ("effect")



Type I error probability

- Suppose we use $\alpha = 0.05$

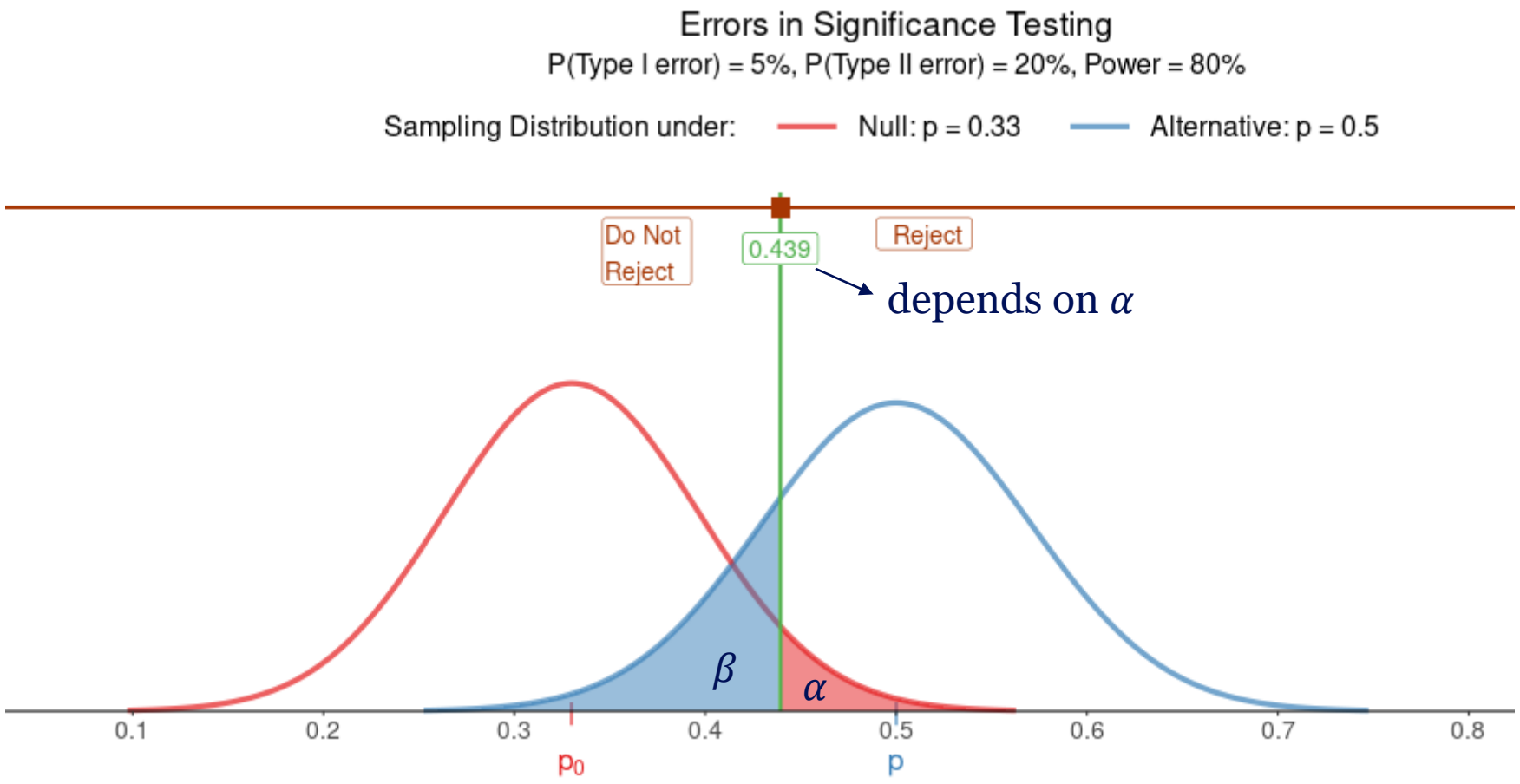
- We reject the null hypothesis if $P \leq 0.05$. P is defined as the probability that the test statistic equals the observed value or even more extreme, **if the null hypothesis is true**.

- So there is a 5% chance of a significant p-value if the hypothesis is true!

- P(type I error) = α

If P(type I error) = α…

- Why not always make α super small?
- https://istats.shinyapps.io/power/



- If we decrease α, P(type I error) goes down, but β=P(type II error) goes up!
- Always report the P-value, not just "reject H_0"
  - Are $p = 0.049$ and $p = 0.051$ really substantially different?

Type II error probability

- β = P(type II error) depends on the true value of the parameter, and its distance to the value under the null hypothesis

- The farther the true parameter value falls from the null value, the **easier** it is to reject the null hypothesis, and P(type II error) goes down

- Power of test = $1 - \beta$ = P(reject H_0, given it is false)

- In practice, you want large enough n so that P(type II error) is small for the size of the effect you expect

Limitations of significance tests

- *Statistical significance ≠ practical significance*
  - If mean weight change in anorexia study is +0.1 pounds with p=0.0001, is that significant?

- Significance tests don't tell us about the **effect size** (like a CI does)
- Some tests are "statistically significant" just by chance
  - And some scientific journals only report "significant" results! ☹

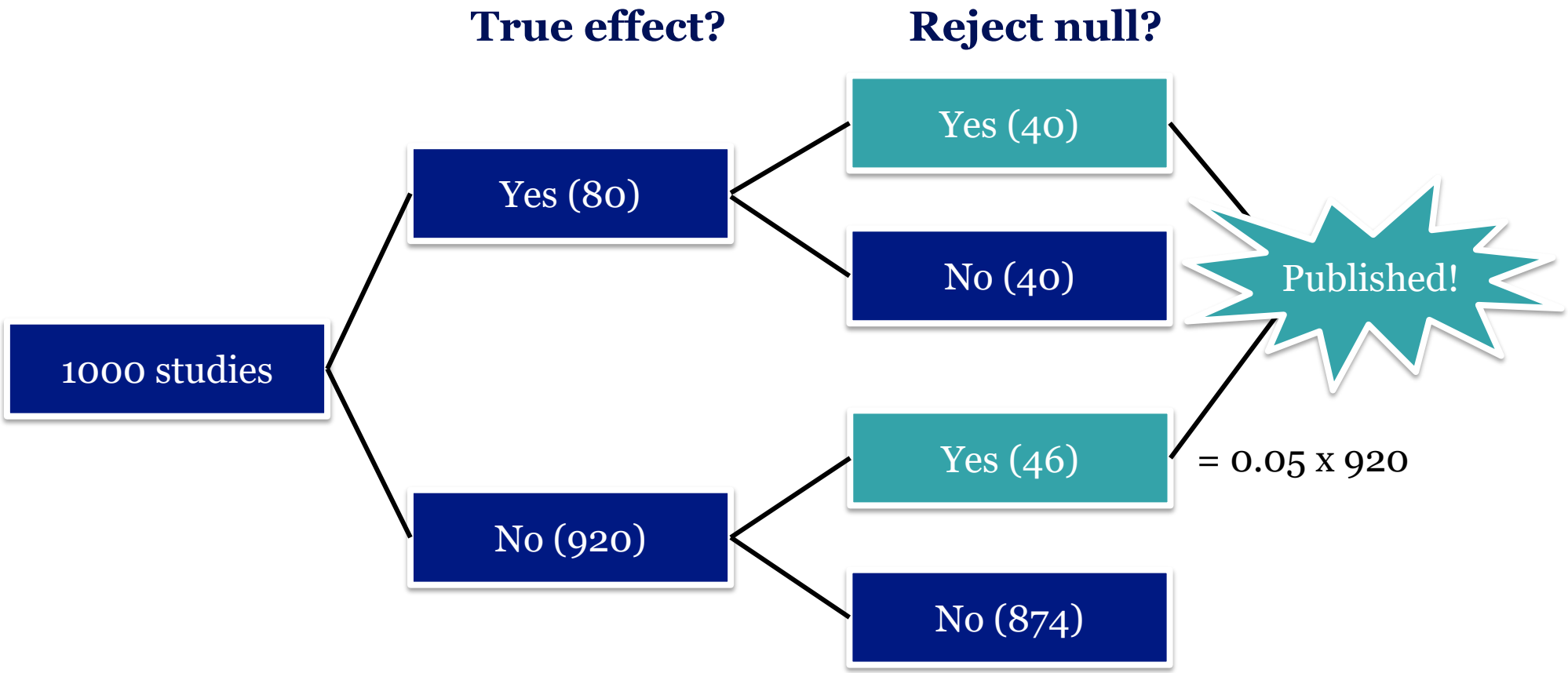
Example: medical "discoveries"

- *Reality*: Most medical studies are "non-significant," not finding an effect
- In medical research, when effects exist but are not strong, they may not be detected with the sample size practical for a study
  - A British Medical Journal article in 2001 estimated that when an effect truly exists, P(Type II error) = 0.50!

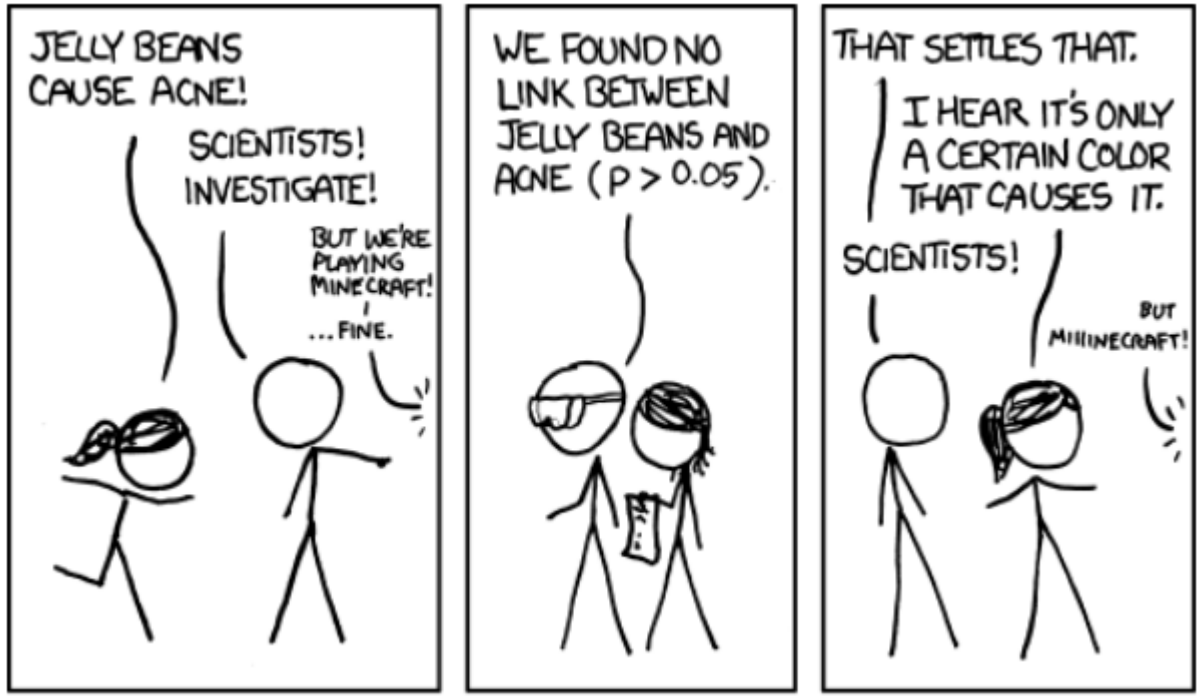
- *In medical studies, suppose an effect actually exists 8% of the time. Could a substantial percentage of medical "discoveries" (i.e., significant results) be Type I errors?*

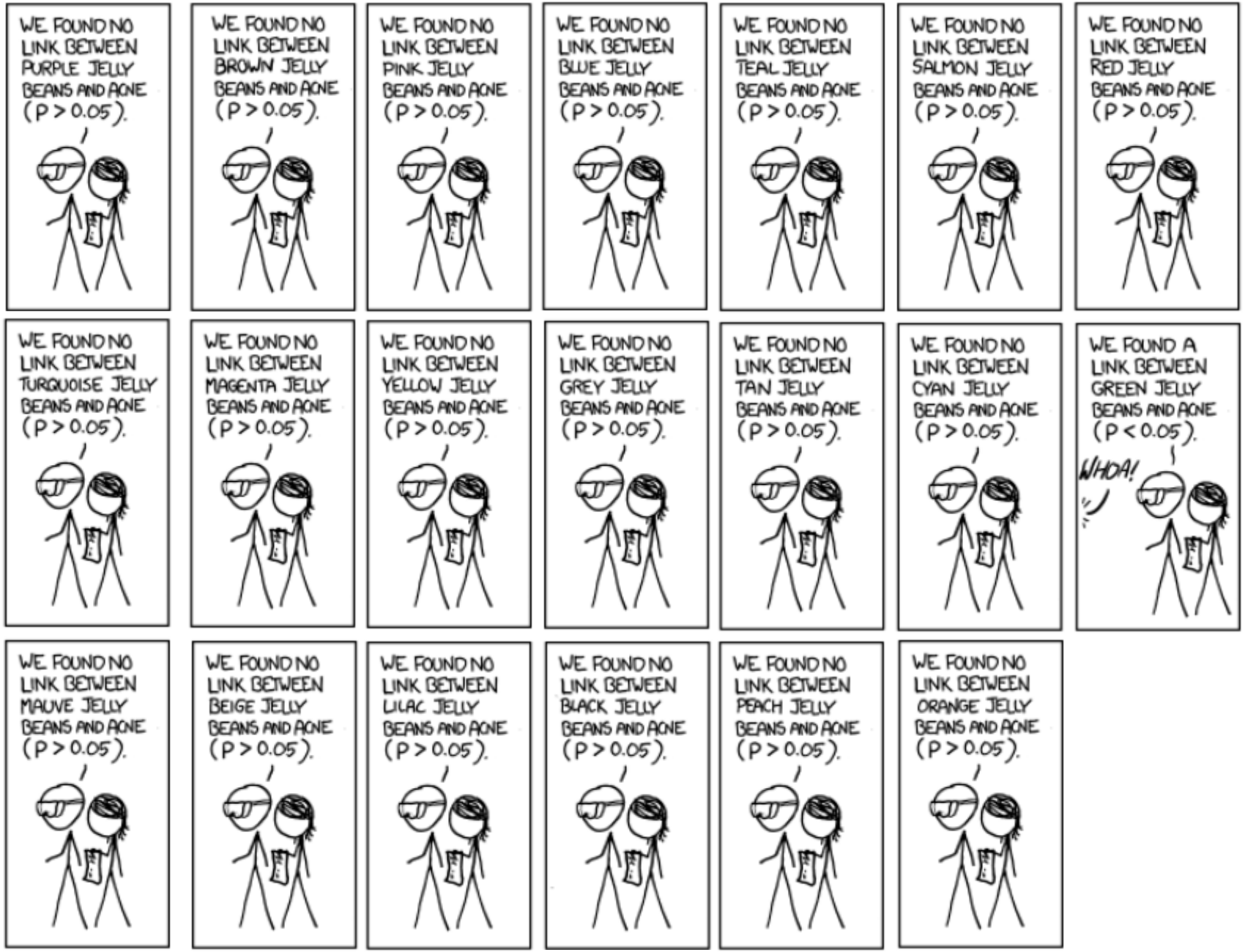
Classified as Internal | Intern

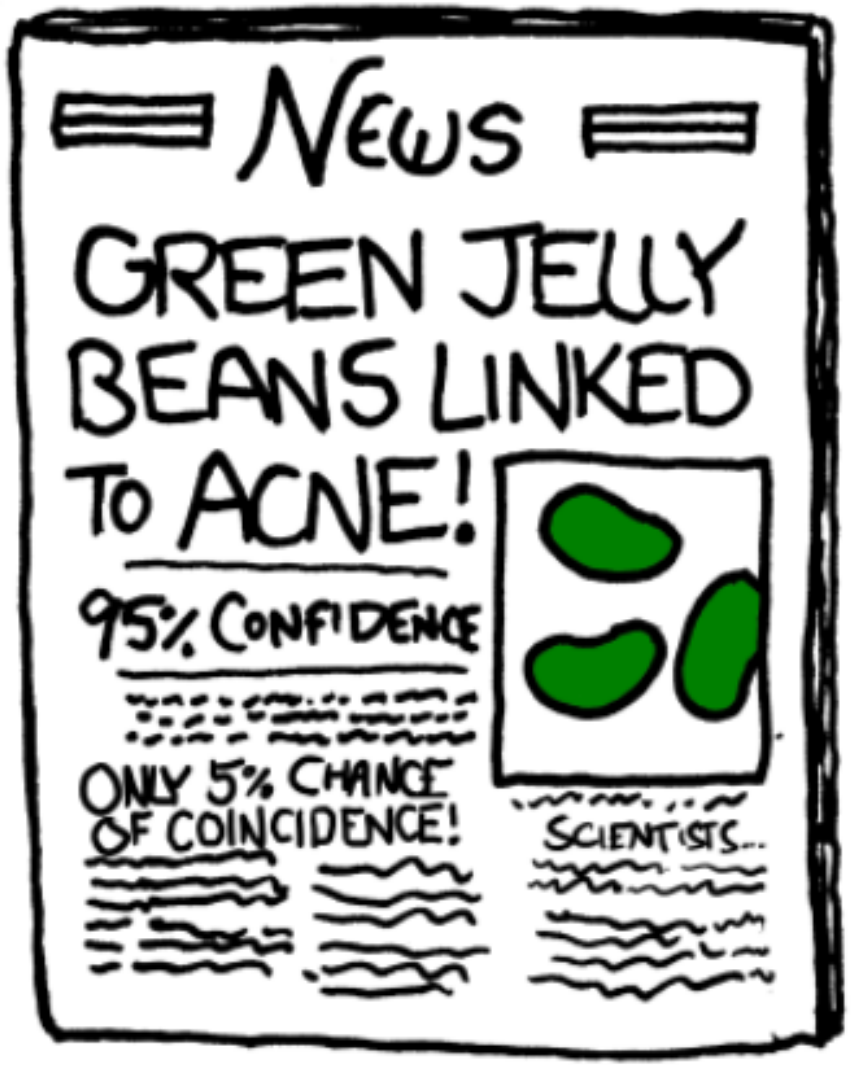
True effect in 8%, all studies use $\alpha = 0.05$



Of studies with rejected H_0, Type I error rate = 46/(46+40) = 0.53


JELLY BEANS CAUSE ACNE!
SCIENTISTS! INVESTIGATE!
BUT WE'RE PLAYING MINECRAFT!
... FINE.
WE FOUND NO LINK BETWEEN JELLY BEANS AND ACNE (P > 0.05).
THAT SETTLES THAT.
I HEAR IT'S ONLY A CERTAIN COLOR THAT CAUSES IT.
SCIENTISTS!
BUT MIIIINECRAFT!


WE FOUND NO LINK BETWEEN PURPLE JELLY BEANS AND ACNE (P > 0.05).
WE FOUND NO LINK BETWEEN BROWN JELLY BEANS AND ACNE (P > 0.05).
WE FOUND NO LINK BETWEEN PINK JELLY BEANS AND ACNE (P > 0.05).
WE FOUND NO LINK BETWEEN BLUE JELLY BEANS AND ACNE (P > 0.05).
WE FOUND NO LINK BETWEEN TEAL JELLY BEANS AND ACNE (P > 0.05).
WE FOUND NO LINK BETWEEN SALMON JELLY BEANS AND ACNE (P > 0.05).
WE FOUND NO LINK BETWEEN RED JELLY BEANS AND ACNE (P > 0.05).
WE FOUND NO LINK BETWEEN TURQUOISE JELLY BEANS AND ACNE (P > 0.05).
WE FOUND NO LINK BETWEEN MAGENTA JELLY BEANS AND ACNE (P > 0.05).
WE FOUND NO LINK BETWEEN YELLOW JELLY BEANS AND ACNE (P > 0.05).
WE FOUND NO LINK BETWEEN GREY JELLY BEANS AND ACNE (P > 0.05).
WE FOUND NO LINK BETWEEN TAN JELLY BEANS AND ACNE (P > 0.05).
WE FOUND NO LINK BETWEEN CYAN JELLY BEANS AND ACNE (P > 0.05).
WE FOUND A LINK BETWEEN GREEN JELLY BEANS AND ACNE (P < 0.05).
WHOA!
WE FOUND NO LINK BETWEEN MAUVE JELLY BEANS AND ACNE (P > 0.05).
WE FOUND NO LINK BETWEEN BEIGE JELLY BEANS AND ACNE (P > 0.05).
WE FOUND NO LINK BETWEEN LILAC JELLY BEANS AND ACNE (P > 0.05).
WE FOUND NO LINK BETWEEN BLACK JELLY BEANS AND ACNE (P > 0.05).
WE FOUND NO LINK BETWEEN PEACH JELLY BEANS AND ACNE (P > 0.05).
WE FOUND NO LINK BETWEEN ORANGE JELLY BEANS AND ACNE (P > 0.05).



https://xkcd.com/882/

Why p-values?

- **Question**
  Why do so many colleges and grad schools teach $p < 0.05$?
- **Answer**
  Because that's still what the scientific community and journal editors use.

- **Question**
  Why do so many people still use $p < 0.05$?
- **Answer**
  Because that's what they were taught in college/grad school.

George Cobb, Professor Emeritus of Mathematics and Statistics,
forum of the American Statistical Association, Feb 2014.

Banning p-values

Nerisa
@neri_peri

Follow

Basic and Applied Social Psychology just went science rogue and banned NHST from their journal. Awesome.
tandfonline.

7:41 PM - 23 Feb 2015

P value ban: small step for a journal, giant leap for science

Editors reject flawed system of null hypothesis testing

BY TOM SIEGFRIED 3:18PM, MARCH 17, 2015

Imagine, if you dare, a world without P values.

Perhaps you're already among the lucky participants in the human race who don't know what a P value is. Trust me, you don't want to. P stands for pernicious, and P values are at the root of all (well, most) scientific evil.

Journal
The American Statistician ›
Volume 73, 2019 - Issue sup1: Statistical Inference in the 21st Century: A World Beyond p < 0.05

Editorial

Moving to a World Beyond "$p < 0.05$"

Ronald L. Wasserstein, Allen L. Schirm & Nicole A. Lazar

Pages 1-19 | Published online: 20 Mar 2019

Download citation https://doi.org/10.1080/00031305.2019.1583913 Check for updates

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EDITORIAL

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Moving to a World Beyond "$p < 0.05$"

“Statistically significant”

Read, interpret and use as

statistically noticeable

or, similarly,

statistically detectable

(No significance hence means there is no detectable effect, which is true!)

This course vs practice

- In this course we often use numeric examples, where we have to decide on ‘significance’ without context

- In practice one should **always interpret a p-value in its context**, ideally jointly with other statistics such as an estimated **effect size** and/or a confidence interval

Wrapping up

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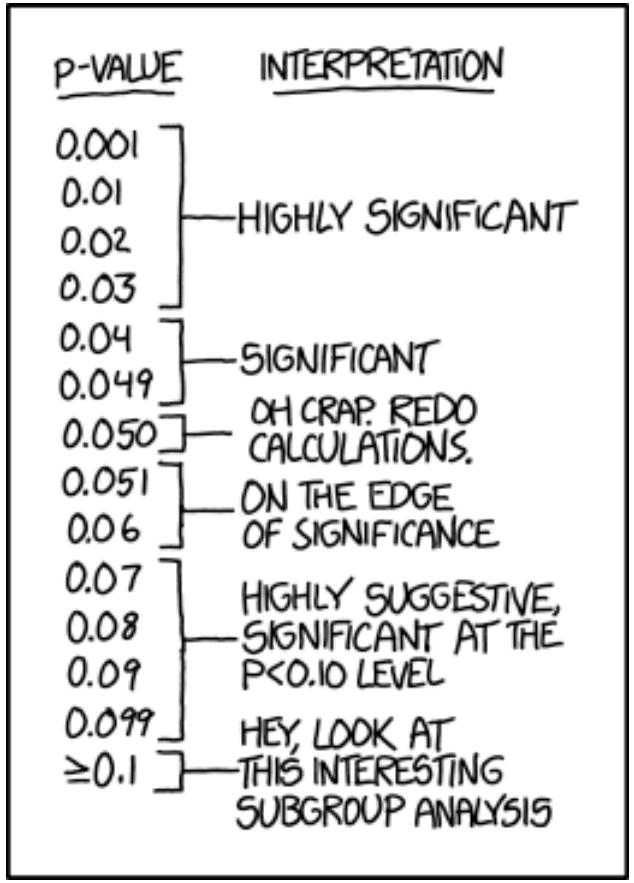
Wrapping up

- Read Chapter 9 of the book

- Today we learned the t-test and z-test
- In the coming weeks we will learn more tests!

- Next week: comparing two groups



https://xkcd.com/1478/